

Edge AI Fall Detection System: A Hardware vs. Software Architecture Comparison

Embedded Systems Mini-Project

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Abstract—Fall detection systems are critical for the safety of elderly individuals and dementia patients. While hardware-accelerated solutions utilizing Field-Programmable Gate Arrays (FPGAs) offer deterministic, real-time performance, they often entail high costs, complex memory management, and prolonged development cycles. This paper presents a software-based Edge Computing alternative using a Raspberry Pi and a pre-trained computer vision model (MediaPipe). We propose a lightweight fall detection pipeline that processes video streams, extracts kinematic skeletal landmarks, and communicates alert states via an I2C-based hardware display. The objective of this mini-project is to provide a Proof of Concept (PoC) and conduct a comparative analysis between the software-based Edge AI architecture and a traditional FPGA/RTL implementation in terms of development agility, hardware complexity, and overall efficiency.

II. SYSTEM ARCHITECTURE

III. METHODOLOGY AND AI ALGORITHM

IV. RESULTS AND PERFORMANCE

V. CONCLUSION

I. INTRODUCTION

The engineering challenge of detecting human falls in real-time is typically addressed using dedicated hardware systems to ensure low latency and high reliability. In our parallel senior design project, a fall detection system is implemented on an Artix-7 FPGA, leveraging custom RTL video pipelines, BRAM memory buffers, and bare-metal SCCB/VGA controllers. While this architecture provides uncompromising deterministic performance, it significantly increases the Time-to-Market (TTM) and engineering overhead.

In modern embedded systems, there is a growing paradigm shift towards Edge AI—running optimized machine learning models on accessible microcomputers. This project investigates the feasibility of replacing the complex hardware pipeline with a software-centric approach. By utilizing a Raspberry Pi and a pre-trained Pose Estimation deep learning model, we aim to extract spatial body coordinates and apply deductive fall-detection logic in Python. Furthermore, to satisfy physical hardware interfacing requirements, the system acts as an I2C master bus, transmitting real-time alerts to a digital display. This paper explores the system architecture, the underlying AI algorithm, and evaluates the trade-offs of Edge Computing versus FPGA acceleration.